Heart Failure: An Approach to a Case of Dyspnoea

Dr. Muhammad Reihan, Internal Medicine | Source: `Reihan_HF.pdf` (49 slides)

Legend: ⭐ = high-yield · *[added]* = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes. The opening-case answer is **reasoned by me; the key is not on the slides**.

🔗 **Bridge from earlier lectures (same lecturer):**

- **CAD: ischaemic heart disease is the commonest cause of HFrEF**; post-MI remodelling.
- **Hypertension:** HTN → LVH → **HFpEF**; ACEi/ARB/β-blockers overlap.
- **Valvular disease:** AS, MR and AR end in HF; **acute MR** (papillary rupture, endocarditis) → de novo acute HF.
- **Arrhythmias:** AF triggers decompensation; **digoxin** (rate control) and its toxicity; ICD/CRT.
- **Pericardial disease:** constrictive pericarditis mimics right HF.
- **Thyroid / pituitary: thyrotoxicosis** and **acromegaly** → high-output HF; thyroid storm → de novo HF.
- **Sleep apnoea:** HF → **Cheyne–Stokes central apnoea** (ASV contraindicated in HFrEF).
- **ARDS lecture:** cardiogenic vs non-cardiogenic oedema (**BNP**, PCWP).

---

1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	⭐ Classify HF by **EF (HFrEF vs HFpEF)**, **ventricle (L/R/biventricular)**, **course (chronic compensated vs acute decompensated)**	Definitions
Knowledge	⭐ **ACC/AHA stages A–D** and **Framingham criteria**	Staging MCQ; 2 major or 1 major + 2 minor
Skill	Apply Framingham to diagnose CHF	
Skill	⭐ Formulate **GDMT** for HFrEF	Which drug ↓ mortality
Skill	⭐ Emergency plan for **acute decompensated HF**	LMNOP-type management

Main subtopics:

- definition and classification
- epidemiology
- pathophysiology and compensation
- clinical features
- ACC/AHA stages
- diagnostics (bloods, biomarkers, imaging, ECG, Framingham)
- management (lifestyle, HFrEF, HFpEF, invasive)
- high-output HF
- acute HF as a medical emergency

---

2. Main Content (Lecture Order)

2.0 Opening Case (Slides 5–6)

Presentation:

- **43-year-old man**, 1 month of progressive dyspnoea
- **two-pillow orthopnoea**, wakes **choking** (PND)
- **AV fistula created 5 months ago** (left upper arm); HTN; **CKD on haemodialysis** (reflux nephropathy)
- **HR 103, bounding pulse**; BP **106/58** (*[added]* wide pulse pressure); RR 22
- **bilateral pitting oedema, JVD**, a **prominent bruit over the fistula**, **bibasal crackles, S3 gallop**

Most likely cause? → ⭐ **F. High-output heart failure** *[reasoning added; key not on slides]*:

- the AV fistula shunts blood past the systemic resistance → **↓ afterload** → compensatory **↑ cardiac output**
- the heart eventually fails: congestion, S3, JVD
- bounding pulse + wide pulse pressure + tachycardia = a **hyperdynamic** circulation
- *[added]* **Nicoladoni–Branham sign:** compressing the fistula → reflex **bradycardia**

Why not the others:

- **dialysis disequilibrium:** neurological, during or right after dialysis
- **sepsis:** no fever
- **PE:** no S3 or congestion
- **constrictive pericarditis:** no S3; Kussmaul's sign, pericardial knock
- **fistula aneurysm:** a local swelling

2.1 Definition and Classification (Slides 7–8) ⭐

- **HF** = a **clinical syndrome** in which the heart **can't supply enough blood to meet the body's demands**.
- Caused by **structural or functional impairment of ventricular filling or ejection**.
- Cardinal **symptoms: dyspnoea, fatigue**. **Signs: oedema, rales**.

Axis	Categories
Ejection fraction	⭐ **HFrEF (systolic)** vs **HFpEF (diastolic)**
Ventricle	**Right** vs **left** vs **biventricular**
Course	**Chronic compensated** vs **acute decompensated**

[added] **EF cut-offs (ACC/AHA 2022):**

Class	LVEF
HFrEF	**≤ 40%**
HFmrEF (mildly reduced)	**41–49%**
HFpEF	**≥ 50%**
HFimpEF (improved)	previously ≤ 40%, now > 40%

The slide's Osmosis diagram uses **normal ≈ 50–70%, borderline 40–50%, systolic HF < 40%**.

2.2 Epidemiology (Slide 9)

- A **leading cause of morbidity and mortality** worldwide.
- Prevalence **~2%** in adults (developed countries); **≥ 10% over age 70**.
- ⭐ **KSA: mean age 57–60 years**, **~10 years younger** than in developed countries.
- Incidence is **lower in women**, but because women live longer they account for **half of all cases**.

2.3 Aetiology (Slide 10) ⭐

Main causes:

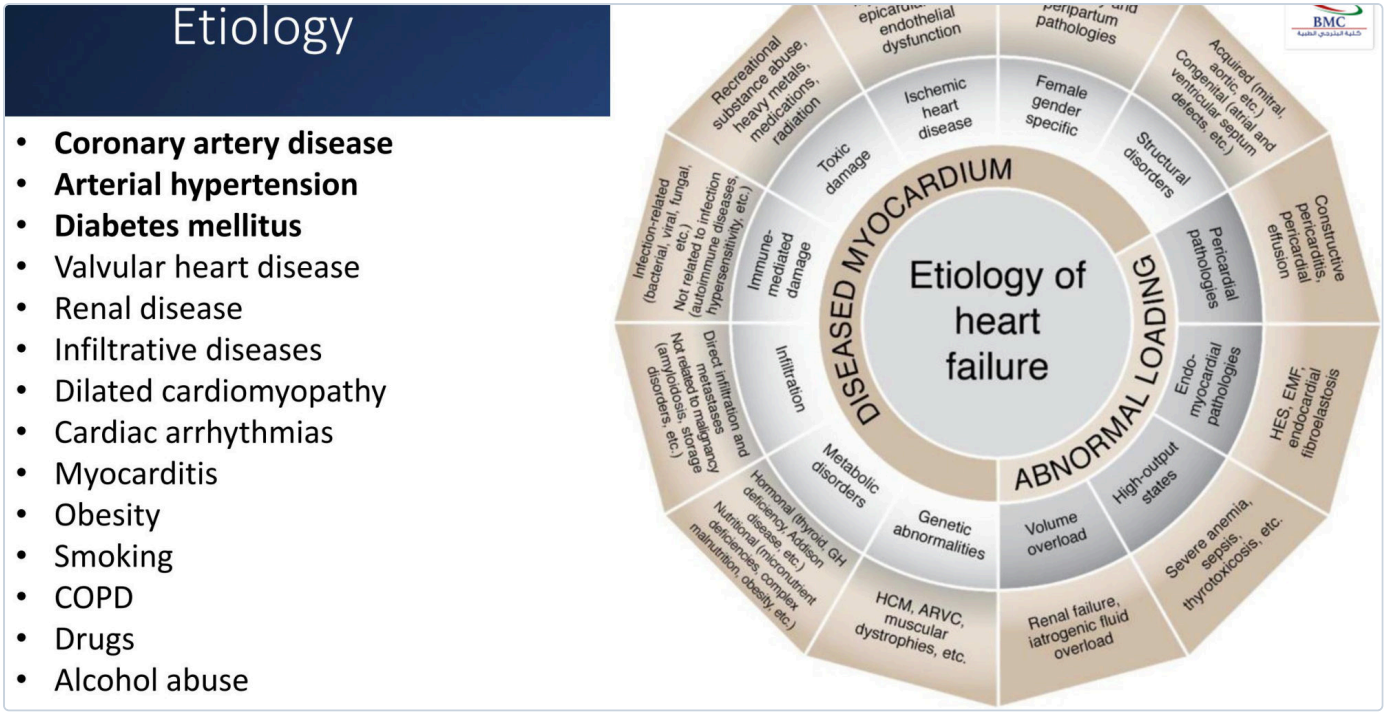
1. ⭐ **CAD**
2. ⭐ **arterial hypertension**
3. ⭐ **diabetes mellitus**

Then:

- valvular heart disease
- renal disease
- infiltrative disease
- dilated cardiomyopathy
- arrhythmias
- myocarditis
- obesity, smoking, COPD
- drugs, **alcohol**

The aetiology wheel splits causes into **diseased myocardium** and **abnormal loading**:

Diseased myocardium	Abnormal loading
Ischaemic	HTN
Toxic (alcohol, drugs, chemo, radiation)	Valvular/structural
Immune/infective (myocarditis)	Pericardial (constriction, effusion)
Infiltrative (amyloid, sarcoid, haemochromatosis)	Endomyocardial (HES, EMF)
Metabolic/hormonal (thyroid, GH, Addison's, nutritional)	**High-output states** (anaemia, sepsis, thyrotoxicosis)
Genetic (HCM, ARVC, dystrophies)	Volume overload (renal failure, iatrogenic)
Peripartum	—

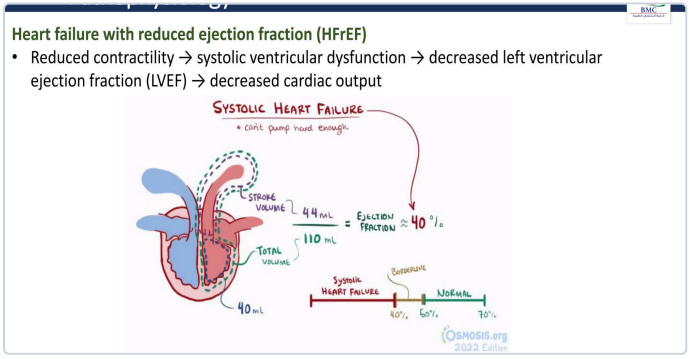


Slide 10: Aetiology of heart failure

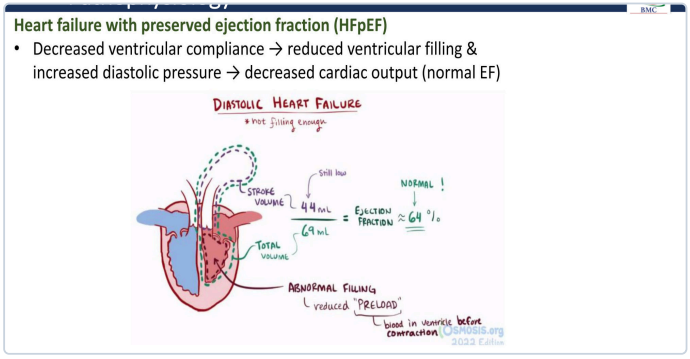
2.4 Pathophysiology (Slides 11–14) ⭐

- **Cardiac output = HR × SV:** 70 beats/min × 70 mL/beat ≈ **4.9 ≈ 5 L/min**.
- **Ejection fraction** = stroke volume ÷ end-diastolic volume. Example: 70/110 ≈ **64%** (normal).

	HFrEF (systolic)	HFpEF (diastolic)
Problem	⭐ ↓ **Contractility**: "can't pump hard enough"	⭐ ↓ **Compliance**: "not filling enough"
Mechanism	Systolic dysfunction → ↓ **LVEF** → ↓ CO	Stiff ventricle → ↓ **filling, ↑ diastolic pressure** → ↓ CO with **normal EF**
Example (slide)	SV 44 / EDV 110 → **EF ≈ 40%**	SV 44 / EDV 69 → **EF ≈ 64%** (SV still low)
[added] Typical patient	Post-MI, DCM, younger man	**Elderly woman with HTN, obesity, DM, AF**
[added] Heart sound	**S3** (volume overload)	**S4** (stiff ventricle, atrial kick)



Slide 12: HFrEF (systolic)



Slide 13: HFpEF (diastolic)

Forward vs backward failure (slide 14):

- ⭐ **Forward:** ↓ CO → poor organ perfusion → organ dysfunction.
- ⭐ **Backward:**
  - **LV failure** → backs up into the **lungs** (pulmonary congestion)

- **RV failure → systemic venous congestion**

2.5 Compensation and Decompensation (Slides 15–16) ⭐

To maintain CO when stroke volume falls:

1. ⭐ **↑ Adrenergic (sympathetic) activity:** ↑ HR, ↑ contractility, vasoconstriction (↑ afterload)
2. ⭐ **↑ RAAS:** angiotensin II (vasoconstriction), **aldosterone (Na^+/H_2O retention)**, ↑ preload; also **ADH**
3. ⭐ **BNP secretion:** natriuresis and vasodilatation (the "good" counter-regulator)

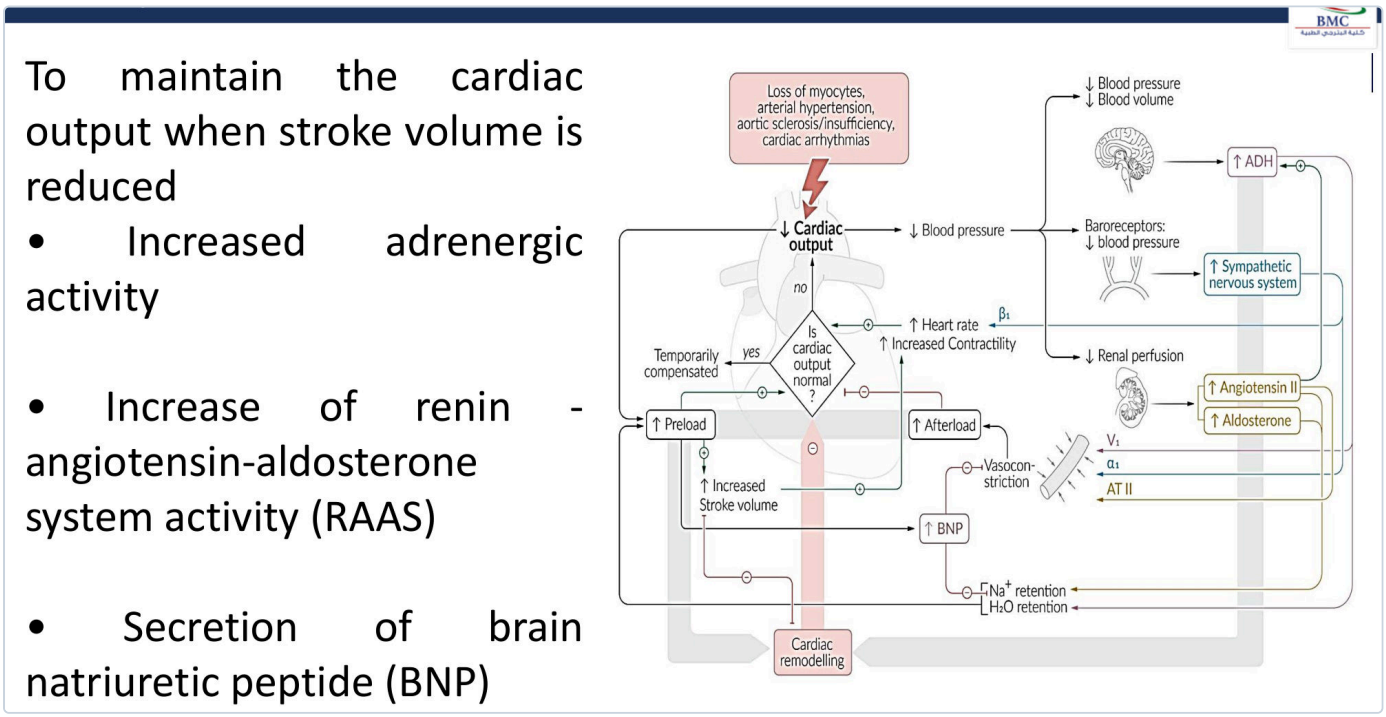
These short-term fixes cause ⭐ **cardiac remodelling** in the long run. That is **why ACEi/ARB/ARNI, β-blockers and MRAs reduce mortality**: they block these maladaptive loops.

Slide 16 summary:

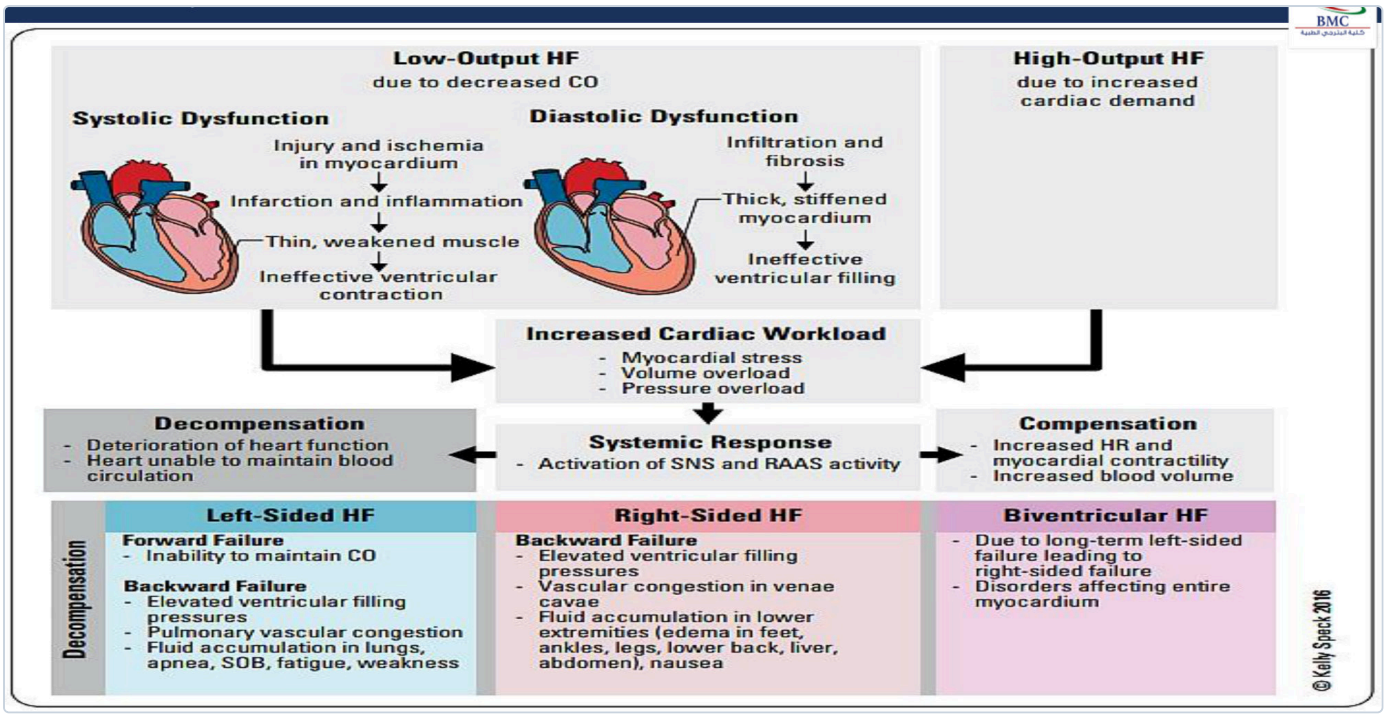
- **Low-output HF:**
  - **systolic dysfunction:** injury/ischaemia, infarction/inflammation → thin, weak muscle
  - **diastolic dysfunction:** infiltration/fibrosis → thick, stiff myocardium
- **High-output HF:** increased cardiac demand.
- All → **↑ workload** (myocardial stress, volume and pressure overload) → **SNS and RAAS activation**.
  - **Compensation:** ↑ HR, ↑ contractility, ↑ blood volume.
  - **Decompensation:** deteriorating function, inability to maintain circulation.

Left-sided HF	Right-sided HF	Biventricular
Forward: can't maintain CO. **Backward:** ↑ filling pressure, **pulmonary congestion** → apnoea, SOB, fatigue	**Backward:** venous congestion in the venae cavae, **oedema** of feet, ankles, legs, lower back, liver, abdomen; nausea	Long-term LHF → RHF; diseases affecting the whole myocardium

[added] The **commonest cause of right HF is left HF**.



Slide 15: Neurohormonal compensation



Slide 16: Compensation vs decompensation; left vs right vs biventricular

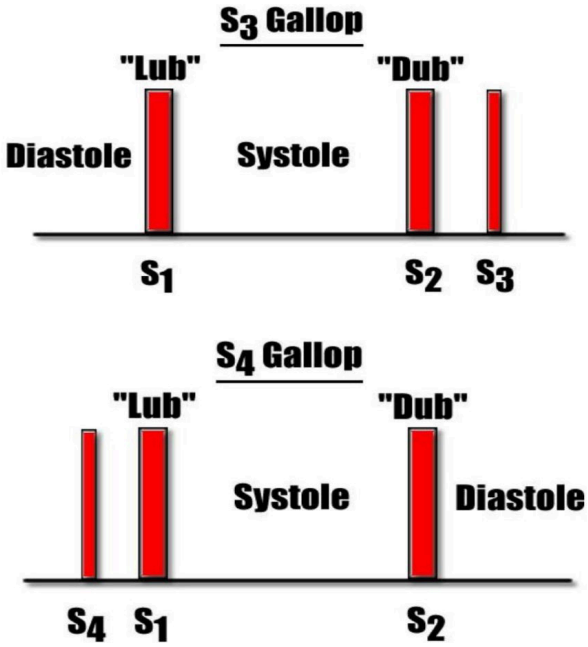
2.6 Clinical Features (Slides 17–19) ⭐

General:

- **nocturia**
- **fatigue**
- **tachycardia**, arrhythmias
- ⭐ **S3 gallop** (early diastolic; volume overload: HFrEF) and **S4 gallop** (late diastolic, before S1; stiff ventricle: HFpEF)
- **cachexia**

Left-sided HF (pulmonary congestion)	Right-sided HF (fluid retention, ↑ CVP)
⭐ **Dyspnoea, orthopnoea, PND**, pulmonary oedema	⭐ **Peripheral pitting oedema**
⭐ **Bilateral basal rales (crackles)**	**Abdominal pain**, nausea, **loss of appetite** (gut and liver congestion)
⭐ **Laterally displaced apex beat** (cardiomegaly)	**Jaundice** (congestive hepatopathy)
Cool, pale extremities (↓ CO)	⭐ **JVD**, ⭐ **hepatojugular reflux**, **hepatosplenomegaly**

- Nocturia
- Fatigue
- Tachycardia
- Various arrhythmias
- S3 gallop
- S4 gallop
- Cachexia



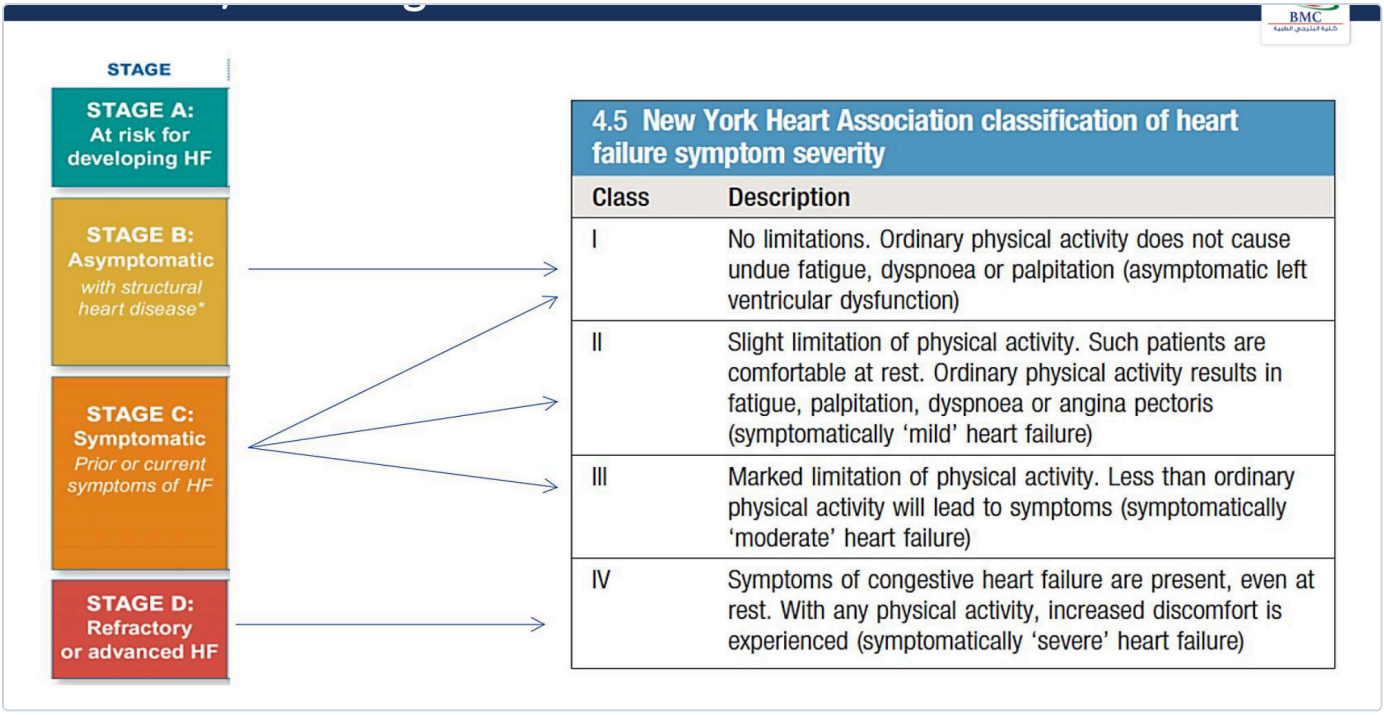
Slide 17: S3 and S4 gallops

2.7 ACC/AHA Stages vs NYHA Class (Slide 20) ⭐⭐

ACC/AHA stage	Description	Maps to NYHA
A	**At risk** for HF (HTN, DM, CAD): **no structural disease, no symptoms**	—
B	**Structural heart disease, asymptomatic** (e.g. LVH, low EF, prior MI)	**I**
C	Structural disease with **prior or current symptoms**	**I–IV**
D	**Refractory / advanced HF**	**IV**

NYHA class	Symptom severity
I	**No limitation**; ordinary activity causes no symptoms
II	**Slight limitation**; comfortable at rest; **ordinary** activity → symptoms (mild HF)
III	**Marked limitation**; **less than ordinary** activity → symptoms (moderate HF)
IV	**Symptoms at rest**; any activity worsens discomfort (severe HF)

⭐ **ACC/AHA stages only progress** (they never go back): a stage C patient who improves stays stage C. **NYHA class changes** with symptoms.



4.5 New York Heart Association classification of heart failure symptom severity

Class	Description
I	No limitations. Ordinary physical activity does not cause undue fatigue, dyspnoea or palpitation (asymptomatic left ventricular dysfunction)
II	Slight limitation of physical activity. Such patients are comfortable at rest. Ordinary physical activity results in fatigue, palpitation, dyspnoea or angina pectoris (symptomatically 'mild' heart failure)
III	Marked limitation of physical activity. Less than ordinary physical activity will lead to symptoms (symptomatically 'moderate' heart failure)
IV	Symptoms of congestive heart failure are present, even at rest. With any physical activity, increased discomfort is experienced (symptomatically 'severe' heart failure)

Slide 20: ACC/AHA stages and NYHA classes

🔁 **Recap: Basics.**

- HF = can't meet demand. HFrEF (EF ≤ 40%, can't pump) vs HFpEF (EF ≥ 50%, can't fill).
- Causes: **CAD, HTN, DM**.
- Compensation: SNS + RAAS + ADH (harmful long-term → remodelling) and BNP (helpful).
- Left: orthopnoea, PND, crackles, displaced apex. Right: oedema, JVD, HJR, hepatomegaly. S3/S4.
- Stages A–D (one-way) vs NYHA I–IV (symptoms).

▶️ **Up next: Diagnostics.**

2.8 Diagnostics: Bloods and Biomarkers (Slides 21–22) ⭐

Everyone	Selected populations
CBC (anaemia)	**HIV**
BMP: ⭐ **↑ creatinine, ↓ Na^+** (hyponatraemia = poor prognosis)	**Iron studies** (haemochromatosis; *[added]* iron deficiency → IV iron)
Cardiac biomarkers	**ANA**
Glucose/HbA1c	**Lyme serology**
LFTs, **lipids**	**TSH**

⭐ **Natriuretic peptides (BNP or NT-proBNP):**

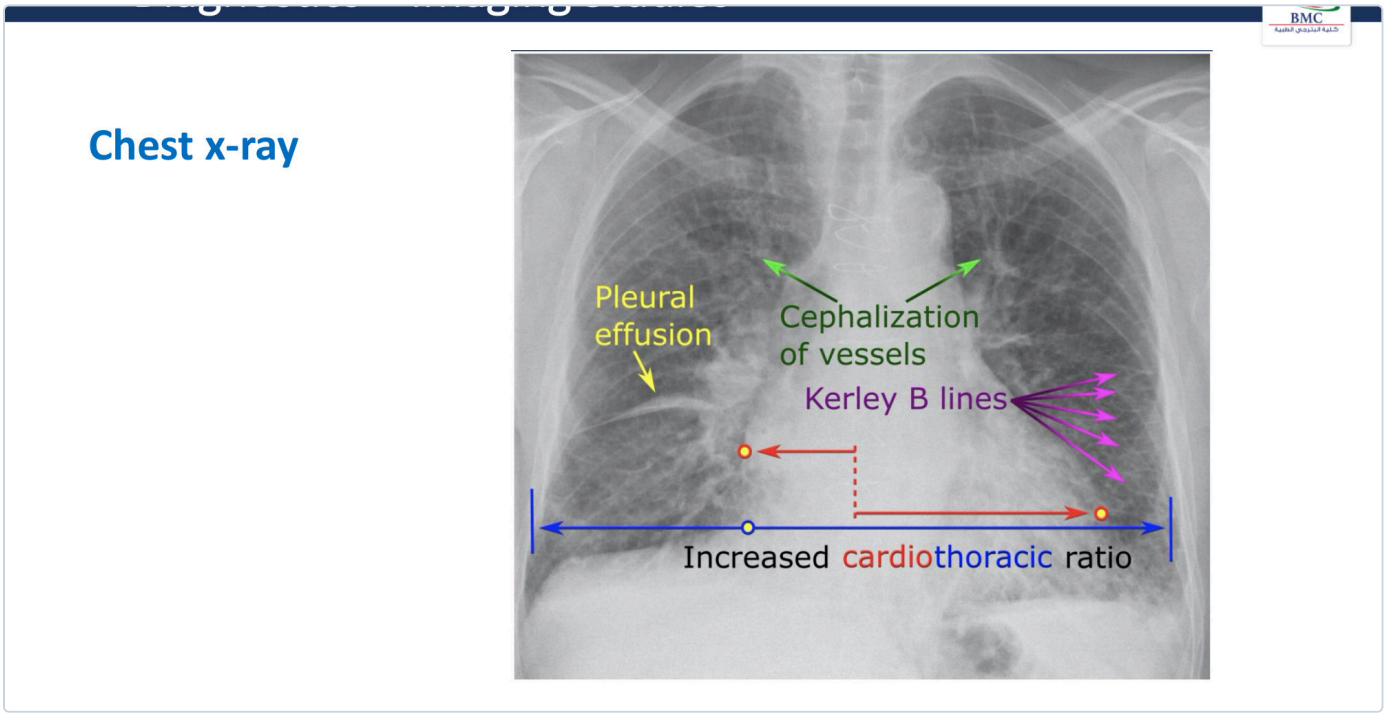
- released in response to **ventricular volume expansion and pressure overload** (wall stretch)
- ⭐ useful to **differentiate HF dyspnoea from COPD**
- *[added]* a **normal BNP effectively rules out HF** (high NPV); cut-offs: BNP < 100 pg/mL, NT-proBNP < 300 pg/mL in acute settings
- ⭐ **↑ with age and kidney dysfunction**
- ⭐ **falsely ↓ in obesity**
- *[added]* also **↑ on sacubitril** (neprilysin breaks BNP down), so use **NT-proBNP** on an ARNI

2.9 Diagnostics: Imaging and ECG (Slides 23–28) ⭐

Test	Role
⭐ **TTE**	**Every patient with suspected new-onset HF**: EF, wall motion, valves, diastolic function
⭐ **CXR**	⭐ **↑ cardiothoracic ratio (> 0.5)**, **upper-lobe diversion ("cephalisation")**, **Kerley B lines**, **pleural effusions** (*[added]* bat-wing alveolar oedema, fissure fluid)
ECG	**Chamber enlargement** (LVH, RVH), **ischaemic heart disease** (Q waves); arrhythmia (AF); *[added]* QRS width/LBBB → CRT
⭐ **Cardiac MRI**	⭐ **Gold standard** for ventricular volume, mass and EF; when TTE is uncertain or for specific diseases (myocarditis, amyloid, sarcoid)
MUGA (radionuclide ventriculography)	EF and volume when other modalities are inadequate or contraindicated

[added] **CXR mnemonic "ABCDE":**

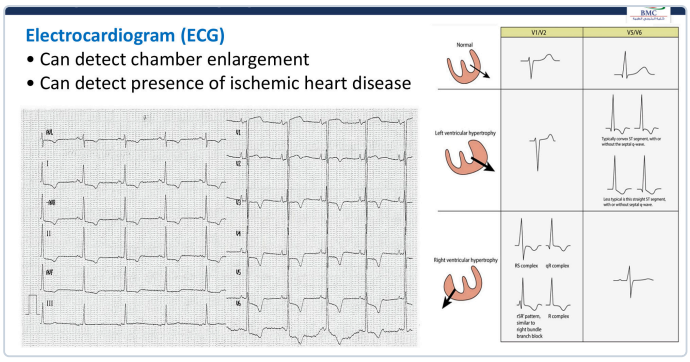
- **A**lveolar oedema (bat-wing)
- Kerley **B** lines
- **C**ardiomegaly
- **D**ilated upper-lobe vessels
- pleural **E**ffusion



Slide 25: CXR of HF (cardiomegaly, cephalisation, Kerley B, effusion)



Slide 24: CXR, normal vs HF



Slide 26: ECG, LVH vs RVH

2.10 Framingham Criteria (Slide 29) ⭐⭐

⭐ CHF = ≥ 2 major, OR 1 major + 2 minor criteria.

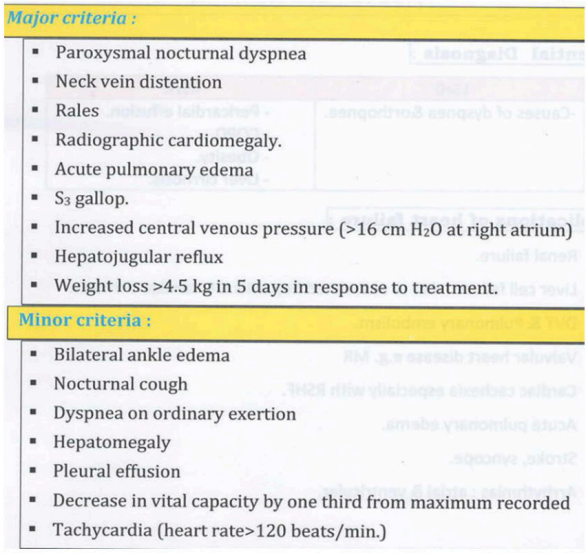
Major criteria	Minor criteria
PND	**Bilateral ankle oedema**
Neck vein distension	**Nocturnal cough**
Rales	**Dyspnoea on ordinary exertion**
Radiographic cardiomegaly	**Hepatomegaly**
Acute pulmonary oedema	**Pleural effusion**
S3 gallop	**↓ Vital capacity by one-third** of the maximum recorded
↑ CVP (> 16 cmH_2O at the RA)	**Tachycardia (HR > 120)**
Hepatojugular reflux	—
Weight loss > 4.5 kg in 5 days in response to treatment	—

[added] Minor criteria count **only if they can't be attributed to another condition** (e.g. pulmonary hypertension, cirrhosis).

Mnemonic *[added]* for the major criteria, "PANCHES + RAW":

- **P**ND
- **A**cute pulmonary oedema
- **N**eck vein distension
- **C**ardiomegaly
- **H**epatojugular reflux
- **E**levated CVP
- **S**3
- **RA**les
- **W**eight loss (> 4.5 kg in 5 days on treatment)

- **Framingham criteria for diagnosis of congestive heart failure:**
- Diagnosis of CHF requires the presence of at least 2 major criteria or 1 major criterion in conjunction with 2 minor criteria.



Major criteria :

- Paroxysmal nocturnal dyspnea
- Neck vein distention
- Rales
- Radiographic cardiomegaly.
- Acute pulmonary edema
- S_3 gallop.
- Increased central venous pressure (>16 cm H_2O at right atrium)
- Hepatojugular reflux
- Weight loss >4.5 kg in 5 days in response to treatment.

Minor criteria :

- Bilateral ankle edema
- Nocturnal cough
- Dyspnea on ordinary exertion
- Hepatomegaly
- Pleural effusion
- Decrease in vital capacity by one third from maximum recorded
- Tachycardia (heart rate>120 beats/min.)

Slide 29: Framingham criteria

🔁 **Recap: Diagnosis.**

- Bloods: CBC, BMP (↑ Cr, ↓ Na), biomarkers, HbA1c, LFTs, lipids, ± TSH, iron, HIV, ANA.
- **BNP:** CHF vs COPD; ↑ with age and CKD; ↓ in obesity.
- **TTE for all**; CXR (CTR > 0.5, cephalisation, Kerley B, effusion); ECG; **cardiac MRI = gold standard**; MUGA.
- **Framingham: 2 major or 1 major + 2 minor.**

▶️ **Up next: Management.**

2.11 Management Overview and Lifestyle (Slides 30–31)

Therapy branches (slide 30):

Branch	Components
Etiologic	Treat the cause
HFrEF	Mortality-improving therapy · symptomatic therapy · lifestyle and comorbidity management
HFpEF	Symptomatic therapy · lifestyle and comorbidities
Invasive	ICD, CRT
Advanced HF	Transplant, LVAD
Acute decompensated HF	Medical emergency

⭐ Lifestyle:

- **sodium restriction (< 4 g/day)**
- **fluid restriction (1.5–2.0 L/day)**
- weight loss; smoking cessation; restrict alcohol
- **exercise programme** (cardiac rehab)
- ⭐ **daily weight monitoring** (*[added]* > 2 kg in 3 days → seek review)

- **immunisation** (influenza, pneumococcal)
- comorbidity management
- ⭐ **avoid medications that worsen HF** (*[added]* **NSAIDs, non-DHP CCBs (verapamil/diltiazem), thiazolidinediones, class I antiarrhythmics, some chemo**)

2.12 HFrEF Pharmacotherapy (Slides 32–34) ⭐⭐

Stage A: treat HTN, hyperlipidaemia, DM. **HF drugs not routinely recommended.**

Stage B (the slide applies this to "everyone with HFrEF"):

Class	Examples	Indication	Effect
⭐ **ACE inhibitors**	Enalapril, ramipril, lisinopril	**Everyone with HFrEF**	**Mortality ↓**
or **ARBs**	Candesartan, losartan, valsartan	**Can't tolerate an ACEi** (cough)	**Mortality ↓**
or ⭐ **ARNI**	**Sacubitril/valsartan**	Can't tolerate ACEi/ARB (slide); *[added]* now **preferred first-line** over ACEi in NYHA II–III	**Mortality ↓**
⭐ **β-blockers**	**Bisoprolol, carvedilol, metoprolol** (*[added]* succinate)	**Everyone with HFrEF**	**Mortality ↓**

Stage C (add):

Class	Examples	Indication	Effect
⭐ **Aldosterone antagonists (MRA)**	**Spironolactone, eplerenone**	**NYHA II–IV and LVEF ≤ 35%**	**Mortality ↓**
⭐ **SGLT2 inhibitors**	**Dapagliflozin, empagliflozin**	**NYHA II–IV**	**Mortality ↓**
Loop diuretics	Torsemide, furosemide	**All patients with fluid retention**	⭐ **Symptoms + ↓ hospitalisation (no mortality benefit)**
Thiazide diuretics	Chlorthalidone, HCTZ	(with loops: sequential nephron blockade *[added]*)	Symptoms
⭐ **Hydralazine + ISDN**	BiDil or separate	**Angiotensin-system blockade intolerance** (*[added]* and **self-identified Black patients** on top of GDMT, A-HeFT)	**Mortality ↓**

Stage D: most patients need **invasive interventions** or a shift to **palliative care**.

Additional therapy (slide 34):

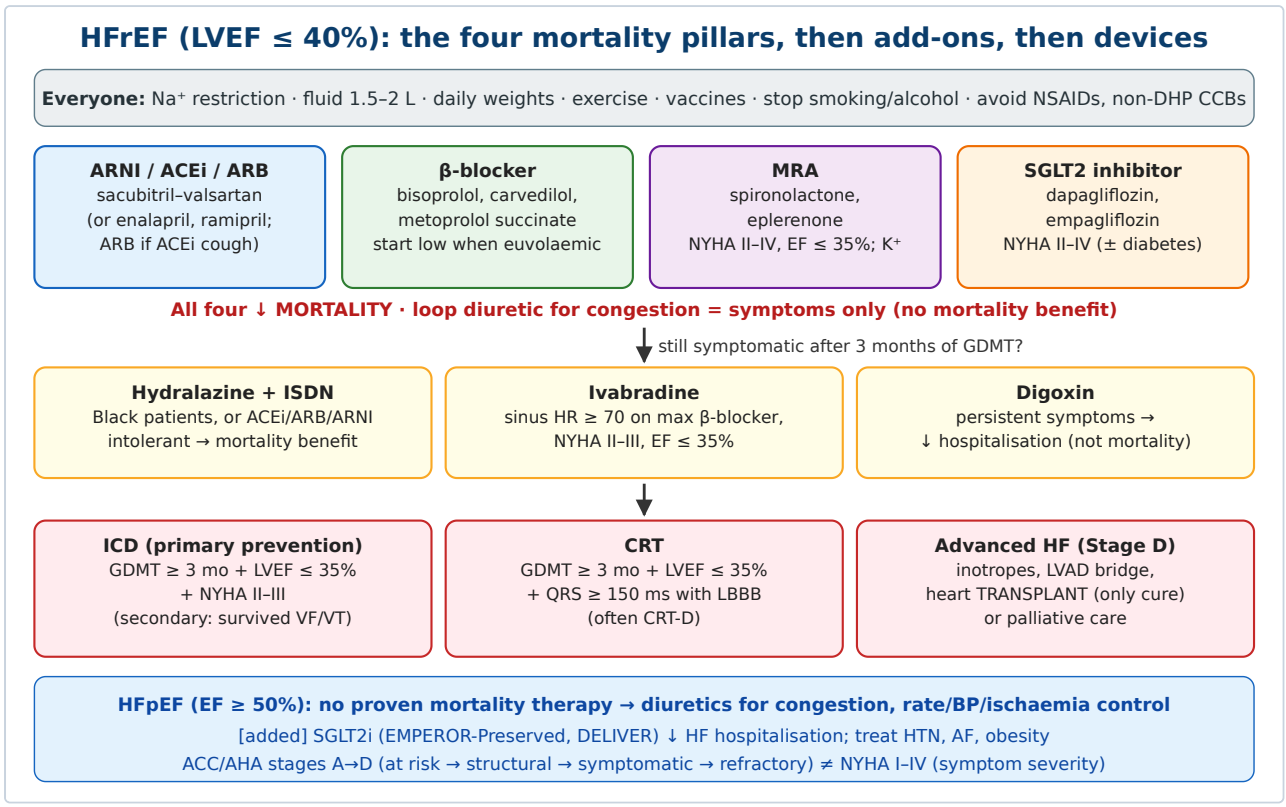
Drug	Indication	Effect
Digoxin	HFrEF with **persistent symptoms despite first-line therapy**	**Symptoms + ↓ hospitalisation** (*[added]* no mortality benefit, DIG trial)
Ivabradine	**All of: NYHA II–III**, stable **HFrEF ≤ 35%**, **sinus rhythm with HR ≥ 70** at rest on the **maximum tolerated β-blocker**	**Symptomatic** (*[added]* ↓ HF hospitalisation; If-channel blocker)

⭐ **The "four pillars"** of HFrEF GDMT *[added framing, matching slides 33–34]*:

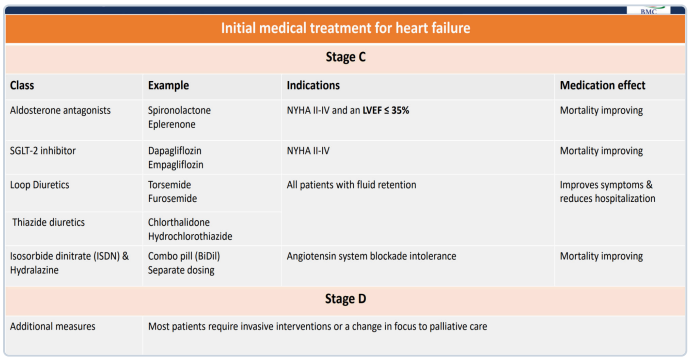
1. **ARNI/ACEi/ARB**

2. **β-blocker**
3. **MRA**
4. **SGLT2i**

All four reduce mortality; start early and titrate.



Added diagram: HFrEF GDMT, the four pillars, add-ons and devices



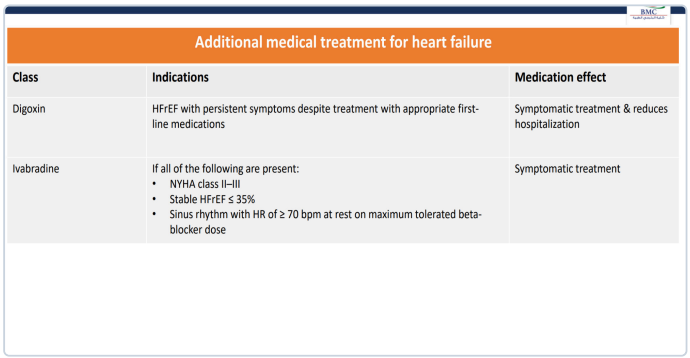
Initial medical treatment for heart failure

Stage C

Class	Example	Indications	Medication effect
Aldosterone antagonists	Spironolactone Eplerenone	NYHA II-IV and an LVEF ≤ 35%	Mortality improving
SGLT-2 inhibitor	Dapagliflozin Empagliflozin	NYHA II-IV	Mortality improving
Loop Diuretics	Torsemide Furosemide	All patients with fluid retention	Improves symptoms & reduces hospitalization
Thiazide diuretics	Chlorthalidone Hydrochlorothiazide		
Isosorbide dinitrate (ISDN) & Hydralazine	Combo pill (BiDil) Separate dosing	Angiotensin system blockade intolerance	Mortality improving

Stage D

Additional measures	Most patients require invasive interventions or a change in focus to palliative care



Additional medical treatment for heart failure

Class	Indications	Medication effect
Digoxin	HFrEF with persistent symptoms despite treatment with appropriate first-line medications	Symptomatic treatment & reduces hospitalization
Ivabradine	If all of the following are present: • NYHA class II–III • Stable HFrEF ≤ 35% • Sinus rhythm with HR of ≥ 70 bpm at rest on maximum tolerated beta-blocker dose	Symptomatic treatment

Slide 33: Stages A and B medical therapy

Slide 34: Stage C and D medical therapy

[added] **GDMT safety:**

- **ACEi/ARB/ARNI:** ↑ K⁺, ↑ creatinine (accept a rise ≤ 30%), angioedema (**36-hour washout ACEi → ARNI**), hypotension; contraindicated in pregnancy and bilateral renal artery stenosis.
- **β-blockers:** start low **when euvolaemic** (not during acute decompensation if hypoperfused); bradycardia.
- **MRA: hyperkalaemia** (avoid if K⁺ > 5.0 or eGFR < 30); **gynaecomastia** with spironolactone (→ eplerenone).
- **SGLT2i:** genital mycotic infection, **euglycaemic DKA**, volume depletion.

2.13 Digoxin and Digitalis Toxicity (Slides 34–35) ⭐⭐

- **Low therapeutic index** → monitor levels.
- ⭐ **Precipitants of toxicity:**

- **old age**
- **renal failure**
- ⭐ **hypokalaemia**, **hypomagnesaemia**, **hypercalcaemia**
- **thyroid disorders**
- *[added]* interacting drugs: **amiodarone, verapamil, quinidine, macrolides**

Clinical picture	Features
GI	⭐ **Anorexia, nausea, vomiting: the 1st symptom**
Neurological	Psychosis/confusion, ⭐ **yellow (xanthopsia) vision**
Cardiac (the most life-threatening)	Arrhythmias: ⭐ **bigeminy/trigeminy**, **AV nodal block**. Any arrhythmia **except Mobitz II** (slide). *[added]* Classic: **paroxysmal atrial tachycardia with block**, **bidirectional VT**, junctional rhythm; hyperkalaemia in acute overdose

Treatment of toxicity:

1. **stop digitalis**
2. **stop diuretics** (especially K^+-losing); correct K^+/Mg^{2+}
3. ⭐ **digoxin-specific antibody fragments (Digibind/DigiFab)**
4. antiarrhythmics: **phenytoin, lidocaine** (ventricular), **amiodarone** (atrial)
   - *[added]* avoid DC cardioversion if possible; **don't give IV calcium**

Prevention:

- lower the dose / drug holidays
- monitor the serum level (slide: **N = 0.5–2 ng/mL**)
- *[added]* the HF target is **0.5–0.9 ng/mL** (higher levels ↑ mortality)

[added] **The digoxin ECG effect** ("reverse tick" / Salvador Dalí moustache: scooped ST depression) indicates **use, not toxicity**.

Management of chronic heart failure

Side effects: digitalis toxicity: It has a low therapeutic index so must be monitored



Precipitating factors increasing toxicity:

(Old age, renal failure, Hypokalemia, Hypomagnesaemia, Hypercalcemia. Thyroid disorders).

Clinical picture of toxicity:-

- **Non-cardiac: GIT**: Anorexia, nausea, vomiting (**1st symptom**) → **Neurological**: Psychosis, **yellow vision**. **Cardiac: (most life-threatening)**: Arrhythmias, (pulses bigeminy and trigeminy), AVN conduction heart block.
- Note: **Any arrhythmia can be a manifestation of digitalis toxicity, except** typeII (mobtiz II) second-degree heart block.

Treatment of toxicity:

- Stop digitalis. → Stop diuretics (especially potassium-losing diuretics) → add Digitalis antibodies (Digibind).
- Anti-arrhythmic drugs (e.g.: phenytoin, lidocaine for ventricular arrhythmias & amiodarone for atrial arrhythmia).
- To avoid toxicity: Decrease the dose &**Drug holiday**. Routine estimation of serum level of digitalis (N=0.5-2ng/ml).

Slide 35: Digitalis toxicity

2.14 HFpEF Management (Slide 36)

- ⭐ **No therapy with proven mortality benefit.**

- ⭐ **Diuretics are the mainstay** for symptom control.
- **Induce bradycardia if tachycardic** (longer diastolic filling) and **relieve ischaemia**.
- Lifestyle and comorbidity management as for HFrEF.
- *[added]* **SGLT2 inhibitors** (EMPEROR-Preserved, DELIVER) **reduce HF hospitalisation** (class 2a); control **BP, AF, obesity** (± GLP-1 RA); MRA/ARNI may help in the lower-EF range.

2.15 Invasive Interventions and Advanced HF (Slides 37–39) ⭐⭐

Cardiomyopathy ↑ the risk of **ventricular arrhythmia and sudden cardiac death**.

Device	Indication
⭐ **ICD: primary prevention** (needs **all 3**)	✓ **GDMT for 3 months** · ✓ **LVEF ≤ 35%** · ✓ **NYHA II–III** (*[added]* expected survival > 1 year; ≥ 40 days post-MI)
⭐ **ICD: secondary prevention**	**Cardiac arrest from VF or VT**
⭐ **CRT**	✓ **GDMT 3 months** · ✓ **LVEF ≤ 35%** · ✓ **QRS > 150 ms with LBBB** (*[added]* NYHA II–ambulatory IV)

Advanced HF (slide 39):

- ⭐ **cardiac transplant is the only cure** for end-stage HF
- **1- and 5-year survival rates are excellent**, but organs are scarce
- **bridging: inotropes** and **mechanical circulatory support** (LVAD)
- **non-candidates → palliative care**

2.16 High-Output Heart Failure (Slide 40) ⭐

- HF secondary to conditions with a **high-output state**: ↑ CO to meet peripheral O_2 demand.
- Can be **physiological (pregnancy)** or **pathological**: **severe anaemia, AV fistula, sepsis, beriberi, thyrotoxicosis**.
- ⭐ **Treatment: correct the underlying aetiology.**

↓ Peripheral vascular resistance (afterload)	↑ Metabolic demand (HR, contractility, afterload)	Mixed (↓ SVR + direct myocardial effect)	Other
Morbid obesity, ⭐ **AV fistula**, **cirrhosis**, erythroderma, **carcinoid**	Myeloproliferative disorders, ⭐ **hyperthyroidism**	⭐ **Sepsis**, **acromegaly**, mitochondrial disease, ⭐ **beriberi (thiamine)**	⭐ **Anaemia**, chronic pulmonary disease

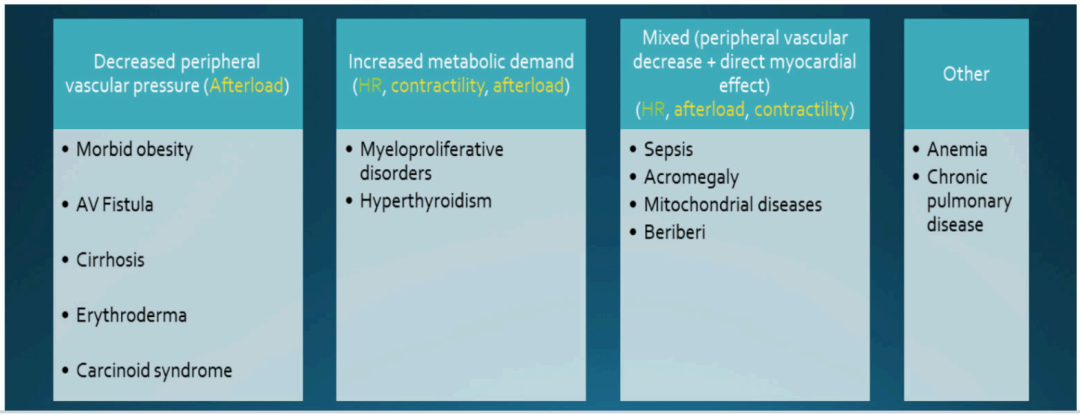
[added] **Clues:**

- warm extremities, **bounding pulse**, **wide pulse pressure**, tachycardia
- **normal or high EF** with high CO on echo

Mnemonic *[added]*: "A FAT PIG":

- **A**naemia
- AV **F**istula
- **A**cromegaly/alcohol
- **T**hyrotoxicosis/thiamine (beriberi)
- **P**regnancy/Paget's
- **I**nfection (sepsis)
- **G**ut (cirrhosis)/carcinoid

- HF secondary to conditions associated with a high-output state
- ↑ CO to meet the peripheral tissue oxygen demands
- Can be physiological (pregnancy)
- Can be pathological (severe anemia, arteriovenous fistula, Sepsis, beriberi & thyrotoxicosis)
- Treatment is directed at **correcting the underlying etiology**



Slide 40: High-output heart failure

🔁 **Recap: Management.**

- Lifestyle: Na < 4 g, fluid 1.5–2 L, daily weights, vaccines, avoid NSAIDs.
- **HFrEF pillars:** ACEi/ARB/**ARNI** + **β-blocker** + **MRA** (EF ≤ 35%) + **SGLT2i** (all ↓ mortality). **Loop diuretic** for congestion. **H-ISDN** if RAAS-intolerant. **Digoxin** and **ivabradine** for symptoms.
- **Digoxin toxicity:** GI first, yellow vision, bigeminy/AV block; ↓ K^+/↓ Mg/↑ Ca and renal failure precipitate it; **Digibind**.
- **HFpEF:** diuretics, rate control (*[added]* SGLT2i).
- **ICD:** EF ≤ 35% + NYHA II–III + 3 months GDMT. **CRT:** + QRS > 150 + LBBB.
- **Transplant** is the only cure.
- **High-output HF** (AV fistula, anaemia, thyrotoxicosis, beriberi, sepsis) → treat the cause.

▶️ **Up next: Acute HF as a medical emergency.**

2.17 Acute Heart Failure (Slides 41–47) ⭐⭐

Type	Definition
⭐ **Acute decompensated HF (ADHF)**	Decompensation of **pre-existing** disease (**most common**)
De novo HF	First presentation, **without known cardiomyopathy**

Triggers of ADHF	Triggers of de novo HF
⭐ **Medication non-compliance** (*[added]* and dietary salt/fluid)	**Acute myocardial dysfunction:** ⭐ **ACS**, **myocarditis**, **drug-induced cardiomyopathy**, ⭐ **peripartum cardiomyopathy**, **thyroid storm**
Ischaemic heart disease	**Acquired valvular pathology:** ⭐ **acute mitral regurgitation**, **bacterial endocarditis**
⭐ **Arrhythmias** (AF)	—
Serious infection/sepsis	—

Triggers of ADHF	Triggers of de novo HF
⭐ **Drugs: NSAIDs, cocaine**	—
Anaemia	—
Renal failure	—
Volume overload	—

Mnemonic *[added]* for ADHF triggers, "FAILURE":

- **F**orgot medication
- **A**rrhythmia/**A**naemia
- **I**schaemia/**I**nfection
- **L**ifestyle (salt, fluid)
- **U**pregulation (thyroid, pregnancy)
- **R**enal failure
- **E**mbolism/**E**ndocarditis

Clinical features	
⭐ **Congestion**	**Acute dyspnoea, orthopnoea**, ⭐ **flash pulmonary oedema**, ↑ work of breathing, **cough** (*[added]* pink frothy sputum), **coarse crackles**
⭐ **Hypoperfusion**	**Weakness, fatigue, altered mental status**, **poor peripheral perfusion** (cold, clammy, oliguria)

[added] **The bedside "warm/cold, wet/dry" profile** (Forrester/Stevenson):

Profile	Meaning	Treatment
Warm–wet	Commonest	**Diuretics ± vasodilators**
Cold–wet	Low CO + congestion → **cardiogenic shock** (next lecture)	**Inotropes** ± vasopressors, then diuresis
Cold–dry	Hypovolaemic / low output	Cautious fluids, inotropes
Warm–dry	Compensated	Optimise GDMT

Differential diagnosis (slide 45):

- ACS
- **pneumonia**
- **COPD/asthma**
- ⭐ **non-cardiogenic pulmonary oedema (ARDS)**
- **PE**
- ⭐ **TRALI** vs ⭐ **TACO** (*[added]* TACO = hydrostatic: **↑ BNP**, HTN, responds to diuretics; TRALI = permeability: **normal BNP**, fever, hypotension)
- **COVID-19**

Diagnostics (slide 46):

- **BNP/NT-proBNP**
- **troponin**
- BMP and electrolytes, CBC, LFTs
- **ECG, CXR, TTE**
- **COVID-19 and flu swab**
- **TSH** (right clinical setting)

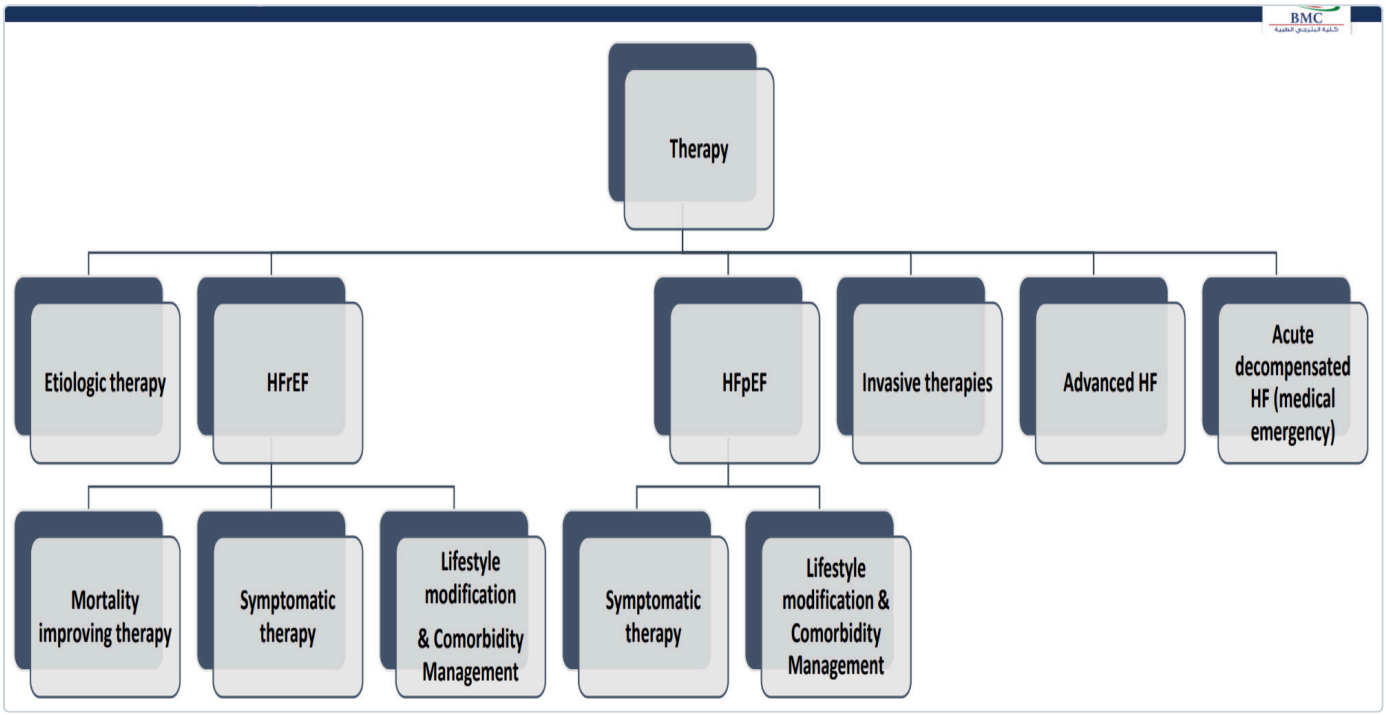
Management (slide 47) ⭐⭐

1. **Respiratory support:**
   - ⭐ **upright position**
   - supplemental **O_2**
   - ⭐ **NIPPV (CPAP/BiPAP)**
   - mechanical ventilation if needed
2. ⭐ **IV loop diuretics:** monitor **urine output and electrolytes**
3. ⭐ **Vasodilators: IV nitroglycerin**, **sodium nitroprusside** (*[added]* if SBP > 110; avoid if hypotensive or with severe AS)
4. **Address the precipitant**
5. Once stable: **optimise GDMT** and **evaluate the aetiology**

Mnemonic *[added]*: "LMNOP":

- **L**asix (furosemide IV)
- **M**orphine (*[added]* now **not routinely recommended**: ↑ mortality signal)
- **N**itrates
- **O**xygen / NIV
- **P**osition (sit upright)

[added] **Hypotensive or hypoperfused (cold-wet) → inotropes (dobutamine, milrinone)** ± norepinephrine; see Cardiogenic Shock.



Slide 30: Therapy branches

> 🔁 **Recap: Acute HF.**
> - ADHF (commonest; non-compliance, ischaemia, AF, infection, NSAIDs/cocaine, anaemia, CKD, volume) vs de novo (ACS, myocarditis, peripartum, thyroid storm, acute MR, IE).
> - Congestion vs hypoperfusion.
> - DDx: ARDS, TRALI/TACO, PE, pneumonia, COPD.
> - **Sit up, O_2/NIV, IV loop diuretic, IV nitrates, treat the trigger, then GDMT.**

---

3. High-Yield Comparison Tables

Table A: HFrEF vs HFpEF

	HFrEF	HFpEF
EF	**≤ 40%**	**≥ 50%**
Defect	**Contractility** (systolic)	**Compliance/relaxation** (diastolic)
Ventricle	**Dilated**, thin	**Thick (LVH)**, small cavity
Typical causes	**CAD/MI**, DCM, alcohol, myocarditis	**HTN**, elderly, female, DM, obesity, AF, **amyloid**
Gallop	**S3**	**S4**
Mortality-reducing drugs	**ARNI/ACEi/ARB, β-blocker, MRA, SGLT2i**; H-ISDN	**None proven** (SGLT2i ↓ hospitalisation *[added]*)
Symptom drugs	Loop diuretic, digoxin, ivabradine	**Diuretics**, rate control
Devices	ICD/CRT if EF ≤ 35%	—

Table B: Left vs Right HF

Left HF	Right HF
Dyspnoea, **orthopnoea, PND**, cough	**Peripheral oedema**, ascites
Basal crackles, pulmonary oedema, effusion	**↑ JVP, hepatojugular reflux**, hepatosplenomegaly
Displaced apex, S3	Anorexia, nausea, RUQ pain, **jaundice**
Cool extremities (low output)	Commonest cause: **left HF**; also cor pulmonale (COPD, OSA, PE)

Table C: ACC/AHA Stage → Therapy

Stage	Therapy
A (at risk)	Treat HTN, lipids, DM; no HF drugs
B (structural, asymptomatic)	**ACEi/ARB(/ARNI) + β-blocker** if EF reduced
C (symptomatic)	**+ MRA + SGLT2i**, loop diuretic, H-ISDN, digoxin, ivabradine; ICD/CRT
D (refractory)	**Transplant, LVAD, inotropes, palliative care**

Table D: Drugs That Help vs Harm in HFrEF

↓ Mortality	Symptoms only	Avoid *[added]*
ARNI/ACEi/ARB, β-blockers (bisoprolol, carvedilol, metoprolol succinate), MRA, SGLT2i, H-ISDN	Loop/thiazide diuretics, digoxin, ivabradine	**NSAIDs**, **verapamil/diltiazem**, **thiazolidinediones** (pioglitazone), **class I antiarrhythmics**, dronedarone, saxagliptin

Table E: BNP Interpretation

↑ BNP	↓ BNP (false-low)
HF, **age**, **CKD**, AF, PE, sepsis, *[added]* **ARNI therapy** (BNP, not NT-proBNP)	**Obesity**, *[added]* flash oedema (very early), constrictive pericarditis, mitral stenosis

4. Rules, Exceptions, Patterns

- **CAD, HTN, DM** are the top causes; KSA patients are **~10 years younger** (57–60).
- **S3 = volume overload (HFrEF); S4 = stiff ventricle (HFpEF).**
- **ACC/AHA stages never regress; NYHA class can.**
- **Framingham: 2 major or 1 major + 2 minor.**
- **TTE for every new suspected HF; cardiac MRI is the gold standard for EF.**
- **BNP rules out HF** (CHF vs COPD); ↑ with age and CKD; ↓ in obesity.
- **The four pillars ↓ mortality; diuretics don't** (symptoms and hospitalisation only).
- **MRA needs EF ≤ 35% + NYHA II–IV**; watch K^+.
- **Ivabradine needs sinus rhythm, HR ≥ 70 on a maximal β-blocker**, EF ≤ 35%, NYHA II–III.
- **Digoxin toxicity ↑ with ↓ K^+, ↓ Mg^{2+}, ↑ Ca^{2+}, renal failure, old age, thyroid disease**; GI first, yellow vision, bigeminy/AV block; **Digibind**.
- **ICD primary prevention = 3 months GDMT + EF ≤ 35% + NYHA II–III. CRT = + QRS > 150 + LBBB.**
- **Transplant = the only cure.**
- **High-output HF → treat the cause** (fistula, anaemia, thyroid, beriberi).
- **ADHF commonest trigger = non-compliance**; de novo = ACS, myocarditis, peripartum, acute MR.
- **Acute HF: sit up, O_2/NIV, IV loop, IV nitrates.**

---

5. Clinical Correlations

- **Dialysis patient with an AV fistula, bounding pulse, wide pulse pressure** → high-output HF (opening case).
- **Thyrotoxicosis / thyroid storm** → high-output HF; AF (Thyroid lecture).
- **Beriberi (thiamine deficiency)** in alcohol use → wet beriberi (high-output HF) (Substance Use lecture).
- **Acromegaly** → cardiomyopathy and high output (Pituitary I).
- **Post-MI acute MR / VSD** → de novo pulmonary oedema (CAD lecture).
- **Infective endocarditis** → acute AR/MR → acute HF (IE lecture).
- **Peripartum cardiomyopathy:** last month of pregnancy to 5 months postpartum *[added]* (next lecture).
- **NSAIDs** precipitate ADHF (Na^+/water retention, ↓ diuretic efficacy).
- **COPD vs HF dyspnoea** → BNP; both cause cor pulmonale.
- **Hyponatraemia** in HF = neurohormonal activation and poor prognosis.
- **Iron deficiency** in HFrEF → IV ferric carboxymaltose improves symptoms *[added]*.

---

6. Rapid Revision (Last-Day)

1. **HF** = the heart can't meet demand; structural or functional filling/ejection impairment; dyspnoea, fatigue, oedema, rales.
2. **Classify:** HFrEF/HFpEF · L/R/biventricular · chronic compensated / acute decompensated.
3. **Epidemiology:** ~2% (≥ 10% over 70); KSA mean age **57–60**; women are half of all cases.
4. **Causes: CAD, HTN, DM**, valves, renal, infiltrative, DCM, arrhythmia, myocarditis, obesity, smoking, COPD, drugs, **alcohol**.
5. **CO = HR × SV** (~5 L/min); **EF = SV/EDV**.
6. **HFrEF:** ↓ contractility → ↓ EF. **HFpEF:** ↓ compliance → ↓ filling, normal EF.

7. **Forward failure** = hypoperfusion; **backward failure** = lungs (LV) / systemic veins (RV).
8. **Compensation: SNS, RAAS (+ ADH), BNP** → remodelling.
9. **General features:** nocturia, fatigue, tachycardia, arrhythmias, **S3/S4**, cachexia.
10. **LHF:** dyspnoea, orthopnoea, PND, oedema; basal rales, **displaced apex**, cool extremities. **RHF:** pitting oedema, abdominal pain, jaundice, nausea, anorexia; **JVD, HJR, HSM**.
11. **Stages A** (at risk) · **B** (structural, asymptomatic) · **C** (symptomatic) · **D** (refractory). **NYHA I–IV.**
12. **Bloods:** CBC, BMP (↑ Cr, ↓ Na), biomarkers, HbA1c, LFTs, lipids; ± HIV, iron, ANA, Lyme, TSH.
13. **BNP/NT-proBNP:** stretch; CHF vs COPD; ↑ with age and CKD; **↓ in obesity**.
14. **TTE** for all; **CXR** (↑ CTR, cephalisation, **Kerley B**, effusion); ECG (chambers, ischaemia); **cardiac MRI = gold standard**; MUGA.
15. **Framingham: 2 major or 1 major + 2 minor**. - **Major:** PND, JVD, rales, cardiomegaly, acute pulmonary oedema, S3, CVP > 16, HJR, weight loss > 4.5 kg in 5 days on treatment - **Minor:** ankle oedema, nocturnal cough, DOE, hepatomegaly, effusion, VC ↓ ⅓, HR > 120
16. **Lifestyle:** Na < 4 g, fluid 1.5–2 L, weight loss, stop smoking, limit alcohol, exercise, **daily weights**, vaccines, avoid harmful drugs.
17. **HFrEF mortality drugs:** - **ACEi** (all) / **ARB** (ACEi-intolerant) / **ARNI** (sacubitril–valsartan) - **β-blocker** (bisoprolol, carvedilol, metoprolol) - **MRA** (NYHA II–IV + EF ≤ 35%) - **SGLT2i** (NYHA II–IV) - **H-ISDN** (RAAS-intolerant)
18. **Symptomatic: loop diuretics** (fluid retention), thiazides; **digoxin** (persistent symptoms); **ivabradine** (NYHA II–III, EF ≤ 35%, sinus HR ≥ 70 on max β-blocker).
19. **Digoxin toxicity:** - precipitants: old age, CKD, **↓ K, ↓ Mg, ↑ Ca**, thyroid - features: **GI first**, psychosis, **yellow vision**, bigeminy/trigeminy, AV block; any arrhythmia except Mobitz II - treatment: stop digoxin and diuretics, **Digibind**, phenytoin/lidocaine (ventricular), amiodarone (atrial) - level 0.5–2 ng/mL
20. **HFpEF:** no proven mortality benefit; **diuretics**; slow HR; relieve ischaemia.
21. **ICD primary:** 3 months GDMT + **EF ≤ 35%** + **NYHA II–III**. **Secondary:** VF/VT arrest. **CRT:** 3 months GDMT + EF ≤ 35% + **QRS > 150 + LBBB**.
22. **Transplant = the only cure**; bridges: inotropes, MCS; otherwise palliative care.
23. **High-output HF:** pregnancy; **anaemia, AV fistula, sepsis, beriberi, thyrotoxicosis**; obesity, cirrhosis, carcinoid, acromegaly → treat the cause.
24. **ADHF triggers:** non-compliance, ischaemia, arrhythmia, sepsis, **NSAIDs/cocaine**, anaemia, renal failure, volume. **De novo:** ACS, myocarditis, drug/peripartum CM, thyroid storm, **acute MR**, IE.
25. **Acute HF:** congestion (flash oedema, crackles) + hypoperfusion (confusion, cold peripheries). DDx: ACS, pneumonia, COPD, **ARDS**, PE, **TRALI/TACO**, COVID.
26. **Work-up:** BNP, troponin, BMP, CBC, LFTs, ECG, CXR, TTE, viral swabs, TSH.
27. **Management:** upright, O_2, **NIPPV**, ventilation; **IV diuretics**; **IV nitroglycerin/nitroprusside**; treat the trigger; then GDMT.

---

7. Exam-Style Questions

Q1 (lecture opening case, slides 5–6). A 43-year-old on haemodialysis via an **upper-arm AV fistula created 5 months ago** has orthopnoea, PND, **bounding pulse 103**, BP 106/58, oedema, JVD, a loud fistula bruit, bibasal crackles and an **S3**. Most likely cause? A. AV fistula aneurysm · B. Dialysis disequilibrium · C. Sepsis · D. PE · E. Constrictive pericarditis · **F. High-output heart failure**

> **Answer: F** *[reasoning added; key not on slides]*.
>
> - The fistula ↓ systemic resistance → **↑ CO demand** → eventual congestive failure (S3, JVD, crackles).
> - Bounding pulse and wide pulse pressure = hyperdynamic circulation (slide 40 lists **AV fistula** as a cause).
> - Treatment = correct the cause (fistula banding/ligation).

Q2. A 65-year-old with ischaemic cardiomyopathy (**LVEF 30%**, NYHA II) is on ramipril and furosemide. Which addition will **most reduce mortality**? A. Digoxin · B. Ivabradine · **C. Bisoprolol** · D. Increase furosemide · E. Amlodipine

> **Answer: C** (slide 33). β-blockers ↓ mortality in **everyone with HFrEF**. Then add an **MRA** (EF ≤ 35%) and an **SGLT2i**; consider switching ACEi → **ARNI**. Digoxin, ivabradine and diuretics are **symptomatic** only.

Q3. A woman on digoxin and furosemide for HFrEF presents with nausea, vomiting, confusion and **yellow-tinged vision**. ECG: **ventricular bigeminy**. K^+ 2.9 mmol/L. Which factor precipitated this, and what is the antidote? **A. Hypokalaemia; digoxin-specific antibody fragments** · B. Hyperkalaemia; calcium gluconate · C. Hypocalcaemia; calcium · D. Hypermagnesaemia; dialysis

> **Answer: A** (slide 35).
>
> - **Hypokalaemia** (from the loop diuretic) potentiates digoxin.
> - **Stop digoxin and diuretics, replace K^+/Mg^{2+}, give Digibind.**
> - *[added]* IV calcium is contraindicated.

Q4. A 58-year-old with HFrEF (LVEF 30%), NYHA III after **4 months** of optimal GDMT, has an ECG showing **LBBB with QRS 165 ms**. Best intervention? A. ICD alone · **B. Cardiac resynchronisation therapy (± defibrillator, CRT-D)** · C. Heart transplant · D. Increase β-blocker

> **Answer: B** (slides 37–38).
>
> - CRT criteria: ≥ 3 months GDMT + **EF ≤ 35%** + **QRS > 150 ms with LBBB**.
> - He also meets ICD criteria, hence **CRT-D**.

Q5. A 72-year-old woman with long-standing hypertension has exertional dyspnoea and ankle oedema. Echo: **concentric LVH, LVEF 62%**, impaired relaxation. BNP elevated. Mainstay of therapy? A. Sacubitril–valsartan for mortality benefit · **B. Diuretics for congestion, plus BP and rate control** · C. Digoxin · D. ICD

> **Answer: B** (slide 36).
>
> - **HFpEF:** no therapy with proven mortality benefit.
> - **Diuretics** for symptoms; control **HTN, AF, ischaemia**.
> - *[added]* An **SGLT2i** reduces HF hospitalisation.

---

8. Exam Pitfalls & Traps

Trap	Correct thinking
Furosemide improves survival in HF	❌ **Symptoms and hospitalisation only**; mortality = the four pillars (+ H-ISDN)
Digoxin ↓ mortality	❌ ↓ **hospitalisation** only
ACC/AHA stage improves from C to B after treatment	❌ Stages **never regress**; NYHA class can
NYHA I = stage A	❌ Stage A has **no structural disease**; NYHA applies from stage B/C
β-blocker started at full dose in a wet, hypoperfused patient	❌ Start low **when euvolaemic**; don't start in cardiogenic shock
MRA for every HFrEF patient	⚠️ Slide: **NYHA II–IV + EF ≤ 35%**; check K^+ and eGFR
Ivabradine in AF	❌ Requires **sinus rhythm** with HR ≥ 70
BNP normal in an obese patient excludes HF	⚠️ BNP is **falsely low in obesity**
BNP high = HF	⚠️ Also ↑ with **age, CKD**, PE, AF, sepsis
ICD 2 weeks after starting GDMT	❌ Needs **3 months of GDMT** (and *[added]* ≥ 40 days post-MI)
CRT for any wide QRS	⚠️ Best for **LBBB with QRS > 150 ms**, EF ≤ 35%
HFpEF has a proven mortality drug	❌ None proven (SGLT2i ↓ hospitalisation *[added]*)
Verapamil for rate control in HFrEF *[added]*	❌ Negative inotrope; use β-blocker or digoxin
NSAIDs for joint pain in HF *[added]*	❌ Precipitate decompensation
IV calcium for hyperkalaemia in digoxin toxicity *[added]*	❌ Avoid; use Digibind
High-output HF treated with standard GDMT alone	❌ **Correct the cause** (fistula, anaemia, thyroid, thiamine)
Morphine routinely in acute pulmonary oedema *[added]*	❌ No longer routine; nitrates, diuretics, NIV
Nitrates in hypotensive ADHF *[added]*	❌ Avoid if SBP < 90–100 or severe AS → inotropes